AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 34 · STUDENT LESSON

Capstone Build

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 24: keep intermediate stages inspectable.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of capstone grounded assistant. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of capstone grounded assistant.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 34: Capstone Build

Implementation should preserve **traceability** from requirement to feature.



CREATOR MISSION

Build the smallest complete Study Buddy workflow.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Capstone requirements and corpus. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can explain capstone build using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Capstone Build case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use capstone build to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Build the assistant as a sequence of testable stages: validate request, retrieve authorised **evidence**, decide answerability, draft, validate and return. Avoid merging all errors into one model score.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

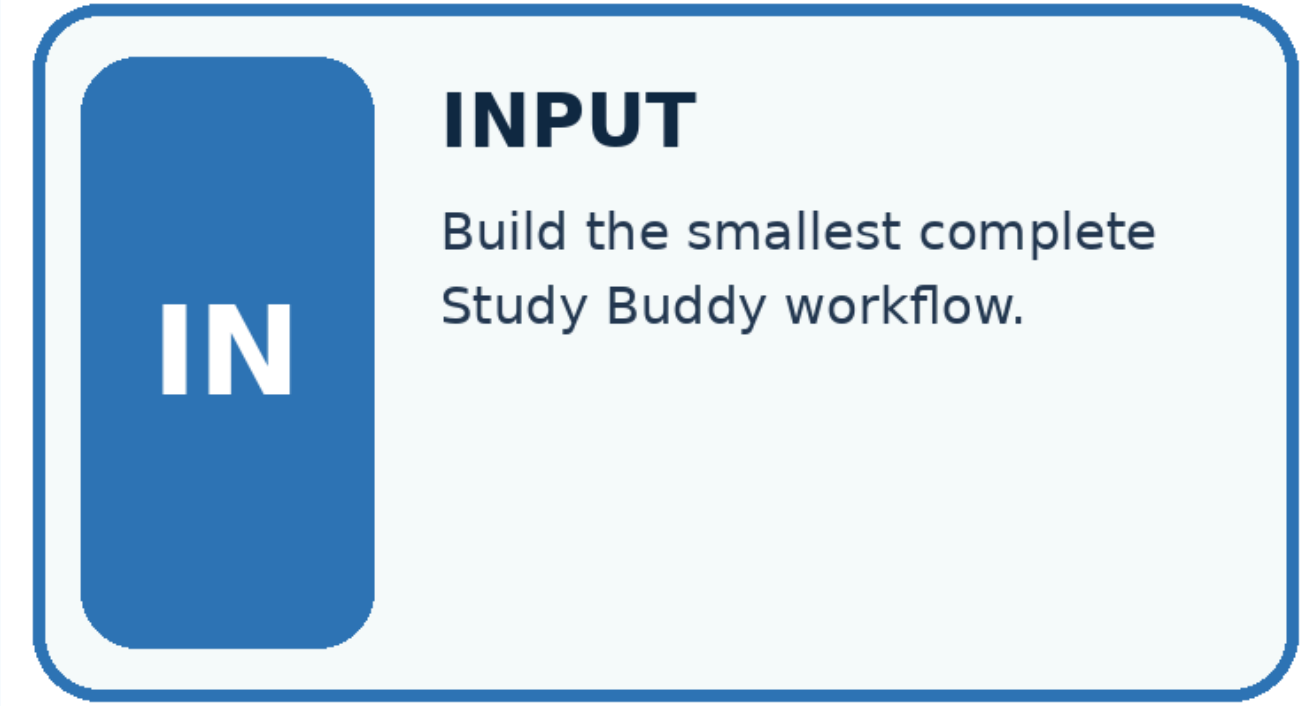
See how it works

> Follow the mechanism for Capstone Build. Connect each labelled input, deciding step and result.

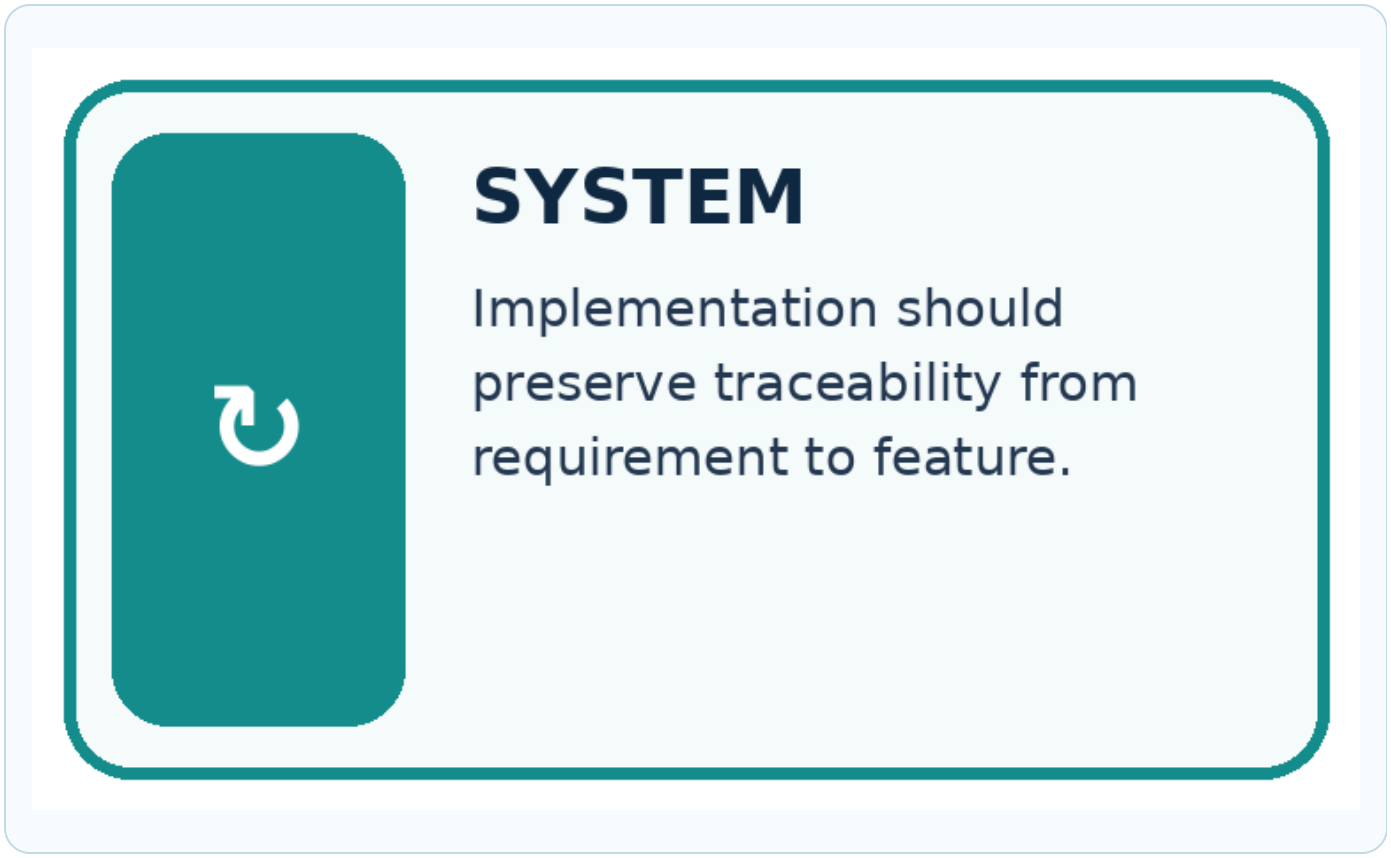
VISUAL MECHANISM

Input → Process → Output

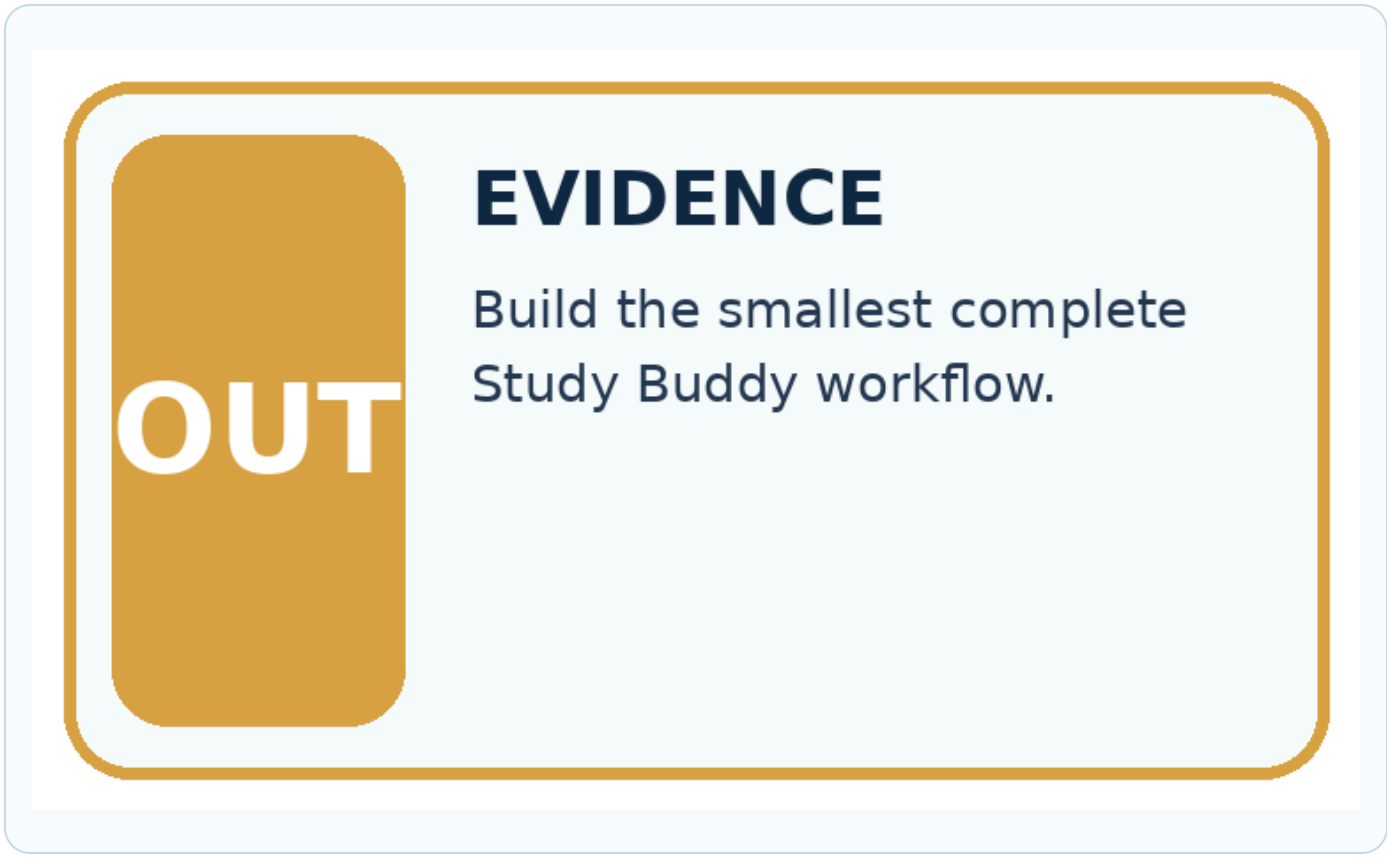
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Implementation should preserve traceability from requirement to feature.

↓


OUT
EVIDENCE
Build the smallest complete Study Buddy workflow.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Capstone grounded assistant — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

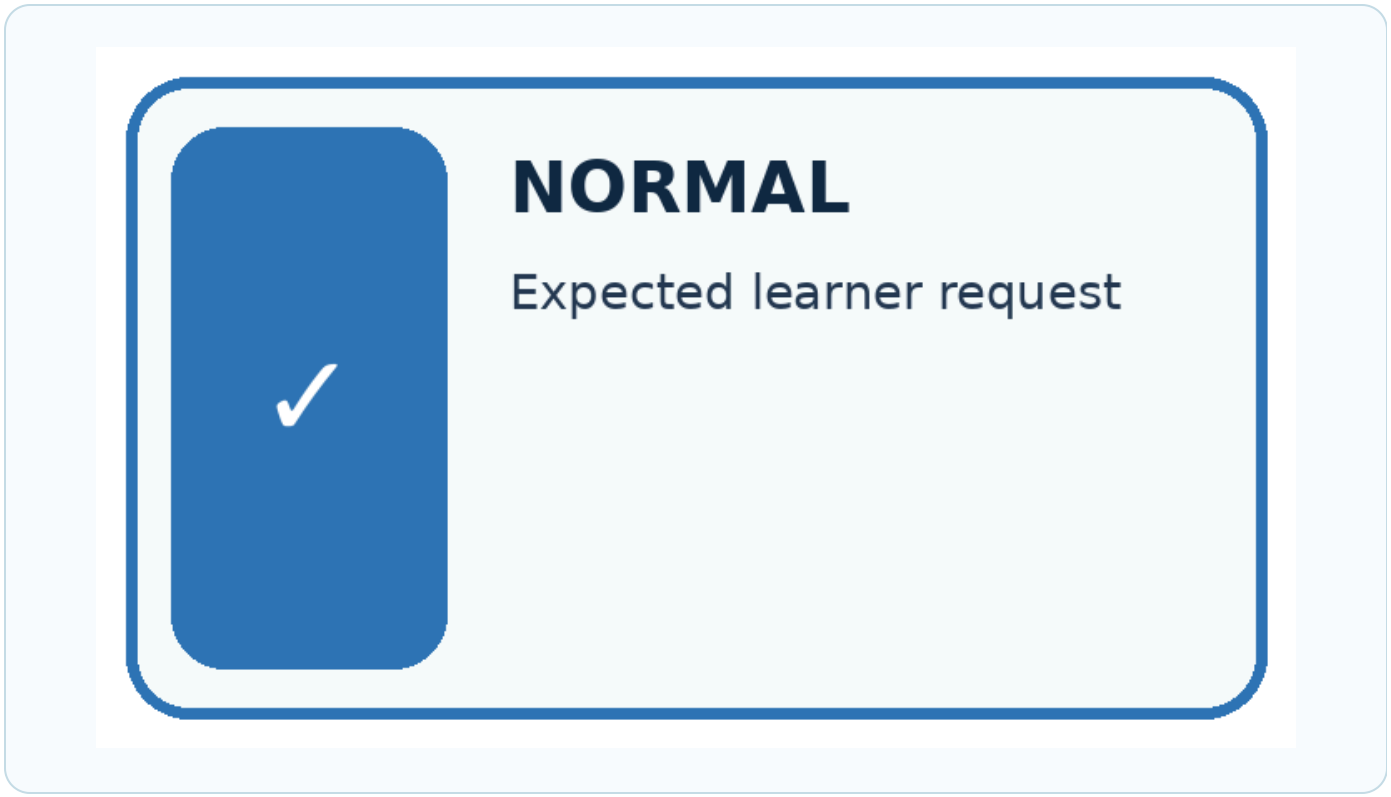
> Compare the supplied cases for Capstone Build. Identify the detail that changes the conclusion.

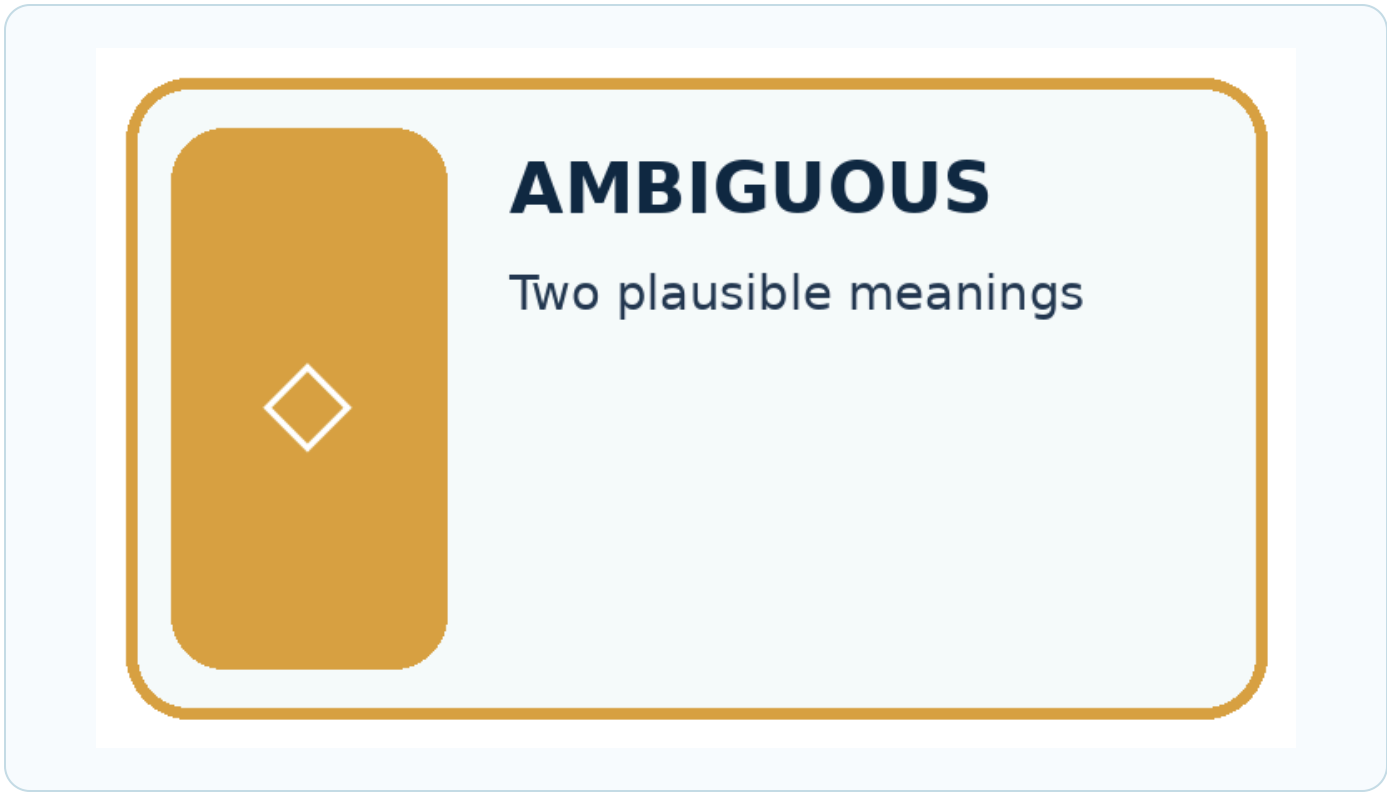
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: E3 supplies losses of 1, 2 and 3 mL.

Predicted result

Prediction: Report those individual losses, not “half” unless starting volumes are also available.

Why the result follows

Reason: Absolute losses alone do not establish each cup’s fractional loss.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Capstone Build to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

TRACEABILITY

link requirement to design, feature, and test

2

INTEGRATION

connect components into one flow





Word	My own example	Not this / boundary
traceability	The ability to connect a result back to its inputs, sources, steps and version.	
integration	Connecting components so that they work together through agreed interfaces.	

Word	My own example	Not this / boundary
version control	A method for recording and comparing changes to identified versions of files.	
minimum complete product	The smallest working version that fulfils its essential user task, including failure handling.	

Words used in the closer explanation

Word	Meaning
Pipeline stage	One operation with defined inputs and failure handling
Answerability gate	A check for sufficient evidence before answering
Semantic validator	A check that claims match task evidence

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone Build, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 34 • WORKED EVIDENCE

1. Requirement R1: answer only from approved sources.
2. Feature: retriever plus source-bound prompt and citation field.
3. Test E4 checks missing evidence triggers abstention.
4. Build one complete path before adding optional interfaces or media.

1. Requirement R1: answer only from approved sources.

2. Feature: retriever plus source-bound prompt and citation field.

3. Test E4 checks missing **evidence** triggers abstention.

4. Build one complete path before adding optional interfaces or media.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

Worked explanation · Capstone Build

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Build

Build a claim checker for a draft summary of a fictional class investigation.

The facts we will use

Supplied facts

Approved record E2: three cups were observed for four days. No measured water loss is provided. Draft says “three cups were observed for four days and all lost half their water”.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/34

Capstone Build

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract claims

Separate cup count, observation duration and the claimed water loss.



2. Match evidence

E2 supports three cups and four days; it supplies no water-loss result.



3. Repair draft

Retain the supported procedure facts; state that water-loss results were not supplied.

CHANGE ONE THING

E3 supplies losses of 1, 2 and 3 mL.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Extract claims	Separate cup count, observation duration and the claimed water loss.
2. Match evidence	E2 supports three cups and four days; it supplies no water-loss result.
3. Repair draft	Retain the supported procedure facts; state that water-loss results were not supplied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

The examples are invented **observation**s; no real classroom investigation or model run is claimed.

Keep the stated limit

Read the labelled picture: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: E3 supplies losses of 1, 2 and 3 mL. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone Build claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: More features create a more complete first build. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

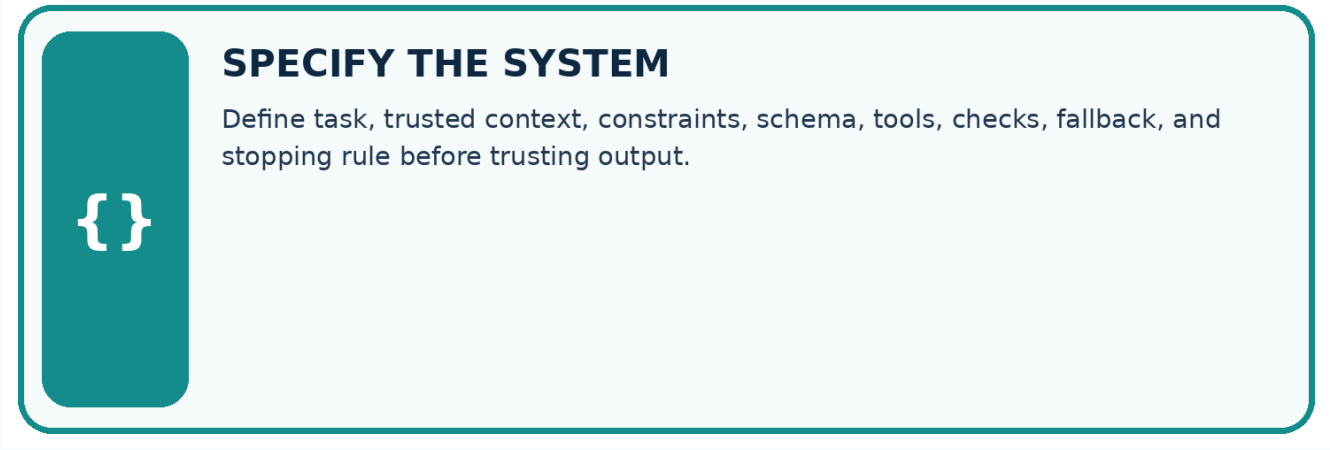
Practise together

> Practise Capstone Build with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A correct happy path does not establish fallback, access control or side-effect safety.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone grounded assistant”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

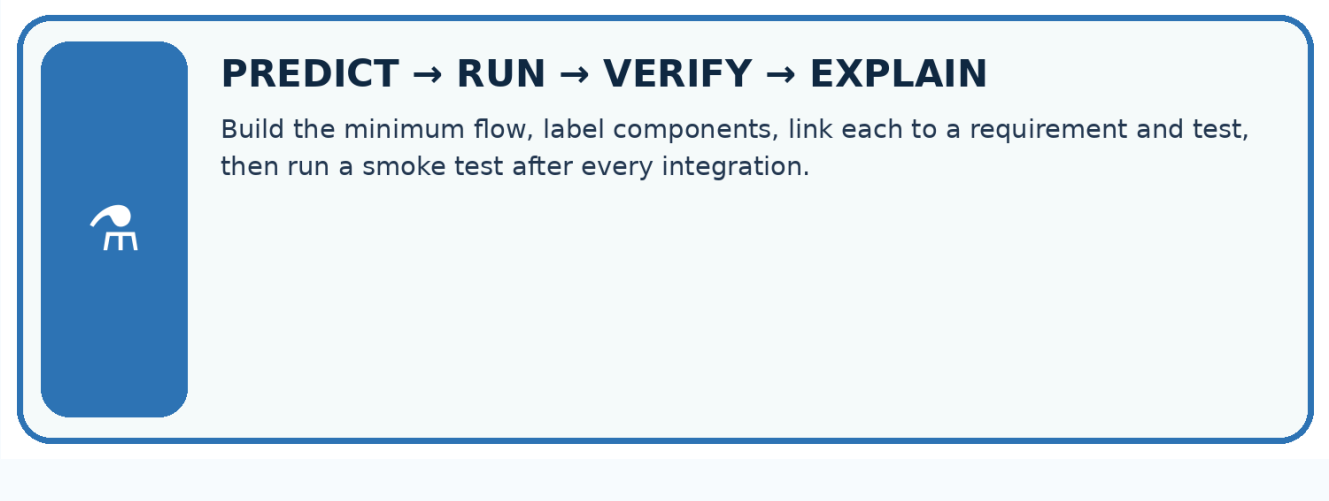
Run the investigation

> Investigate Capstone Build. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Build the minimum flow, label components, link each to a requirement and test, then run a smoke test after every **integration**.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

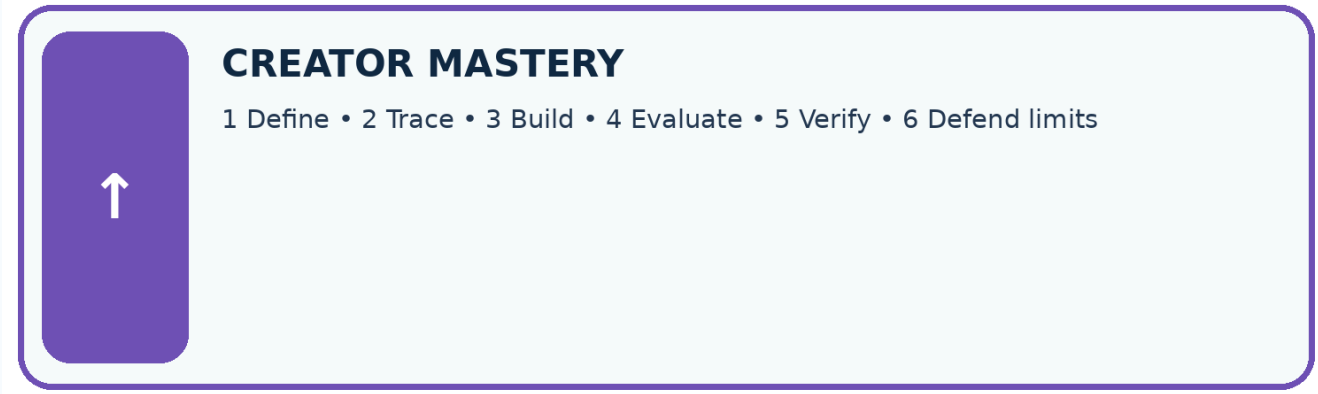
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone Build. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.

VERIFIED STUDY BUDDY

Assemble interface, retrieval, prompt, model/tool, validation, evidence display, and fallback.

THIS WEEK'S CHECKPOINT Assemble interface, retrieval, prompt, model/tool, validation, **evidence** display, and fallback.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

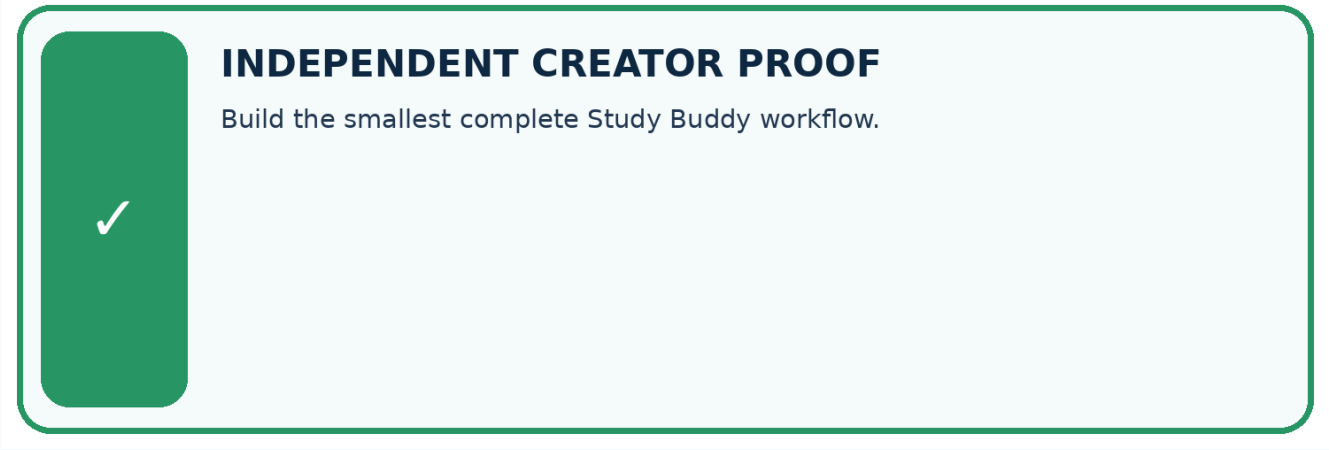
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Build the smallest complete Study Buddy workflow.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Capstone Build case for supported practice. Repeated examples do not become fresh assessments.

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A pipeline retrieves passages, drafts and displays without checking citations.

The question

> What step is missing?

Worked explanation

Validate every factual claim against its citation, plus required fields and boundaries. Revise or stop unsupported answers; retain diagnostic stages.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Validate before displaying the draft

Evidence or stage	Value or result
Retrieve	Relevant approved passages
Draft	Compose the answer
Validate	Check claims and citations
Display or stop	Show supported content or explain the limit

A pipeline retrieves passages, drafts and displays without checking citations.

> What step is missing?

Validate every factual claim against its citation, plus required fields and boundaries. Revise or stop unsupported answers; retain diagnostic stages.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

An experiment note records four fictional seeds observed for six days but supplies no growth results. A generated draft says all four sprouted and the method always works in any soil. Repair it.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> The interface looks complete, but its answer button returns the same saved sentence for every question. What capability is still missing, and how can you expose it?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Assemble interface, retrieval, prompt, model/tool, validation, **evidence** display, and fallback.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone Build, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Build a claim checker for a draft summary of a fictional class investigation.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone grounded assistant	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone Build without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone grounded assistant

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a claim checker for a draft summary of a fictional class investigation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Retain the supported procedure facts; state that water-loss results were not supplied.

1. Extract claims

2. Match **evidence**

3. Repair draft

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair draft Retain the supported procedure facts; state that water-loss results were not supplied. Read the picture in order.

Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: E3 supplies losses of 1, 2 and 3 mL. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Report those individual losses, not “half” unless starting volumes are also available.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report those individual losses, not “half” unless starting volumes are also available. Absolute losses alone do not establish each cup’s fractional loss.

Repair a missing building block

Review Capstone Design (ai-3/33). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Safety Boundaries (ai-3/31). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: If AB7 passes the pattern, may it access teacher-only material?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

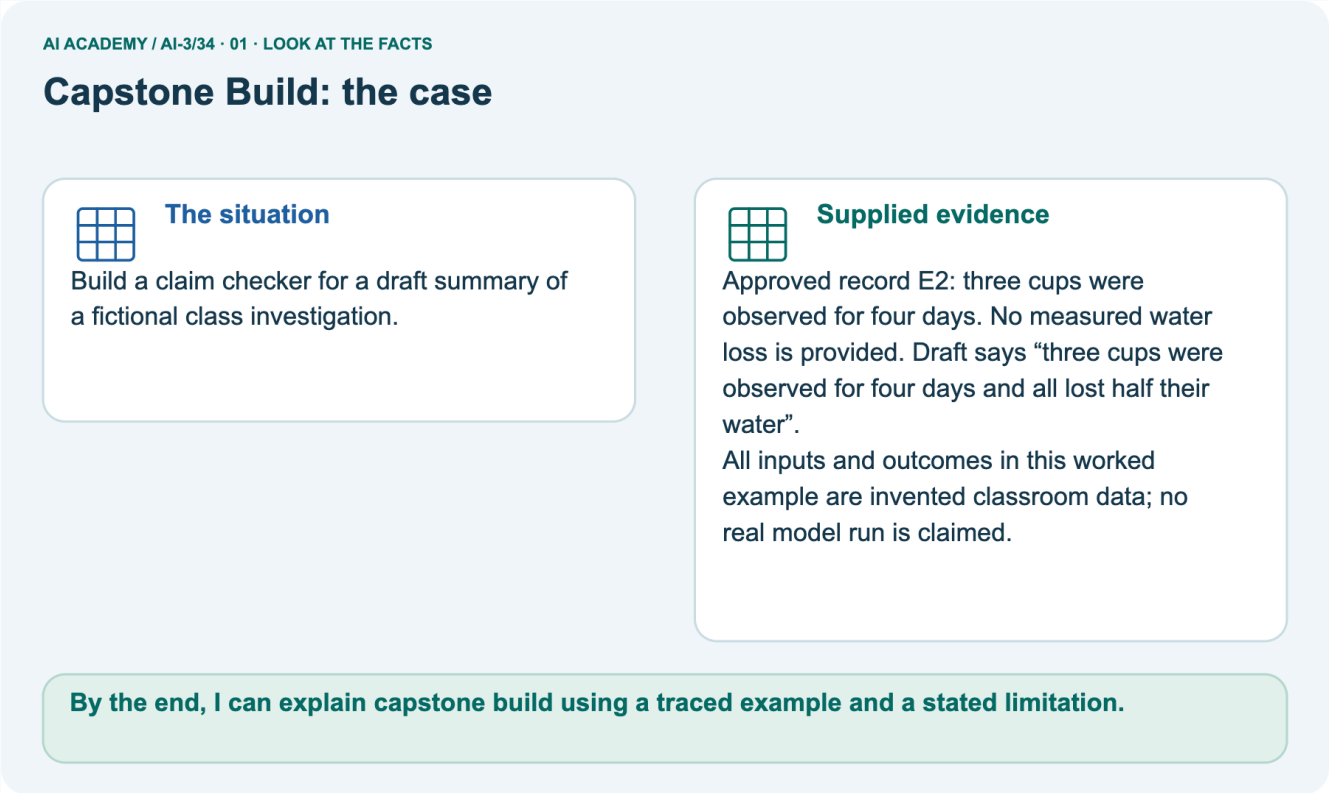
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Capstone Build graphic using its labels, deciding evidence and relevant limitation.

01. Capstone Build: the case



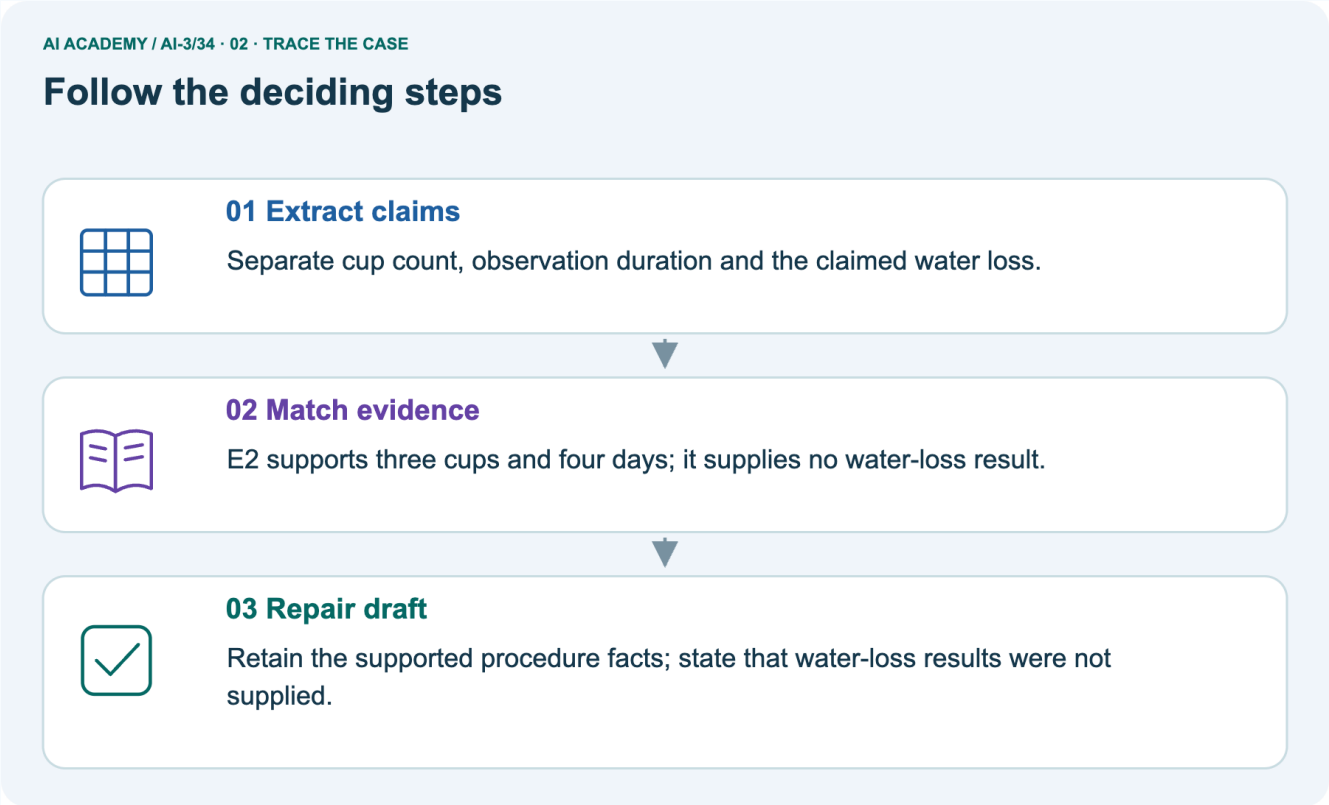
Follow the labelled picture with the matching explanation. Open the vector diagram

Build a claim checker for a draft summary of a fictional class investigation. Supplied facts: Approved record E2: three cups were observed for four days. No measured water loss is provided.

Draft says “three cups were observed for four days and all lost half their water”. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

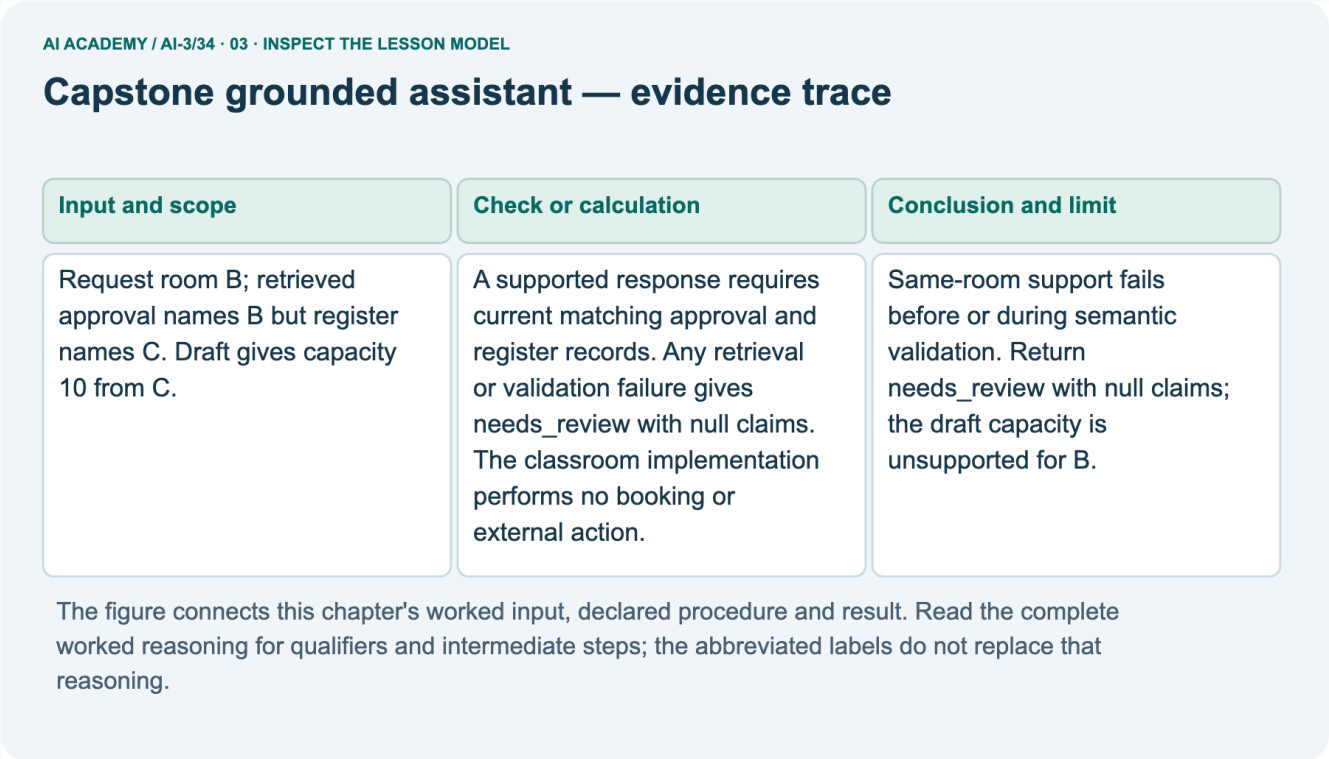


Follow the labelled picture with the matching explanation. Open the vector diagram

Extract claims: Separate cup count, **observation** duration and the claimed water loss. → Match **evidence**: E2 supports three cups and four days; it supplies no water-loss result. → Repair draft: Retain the supported procedure facts; state that water-loss results were not supplied.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone grounded assistant — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

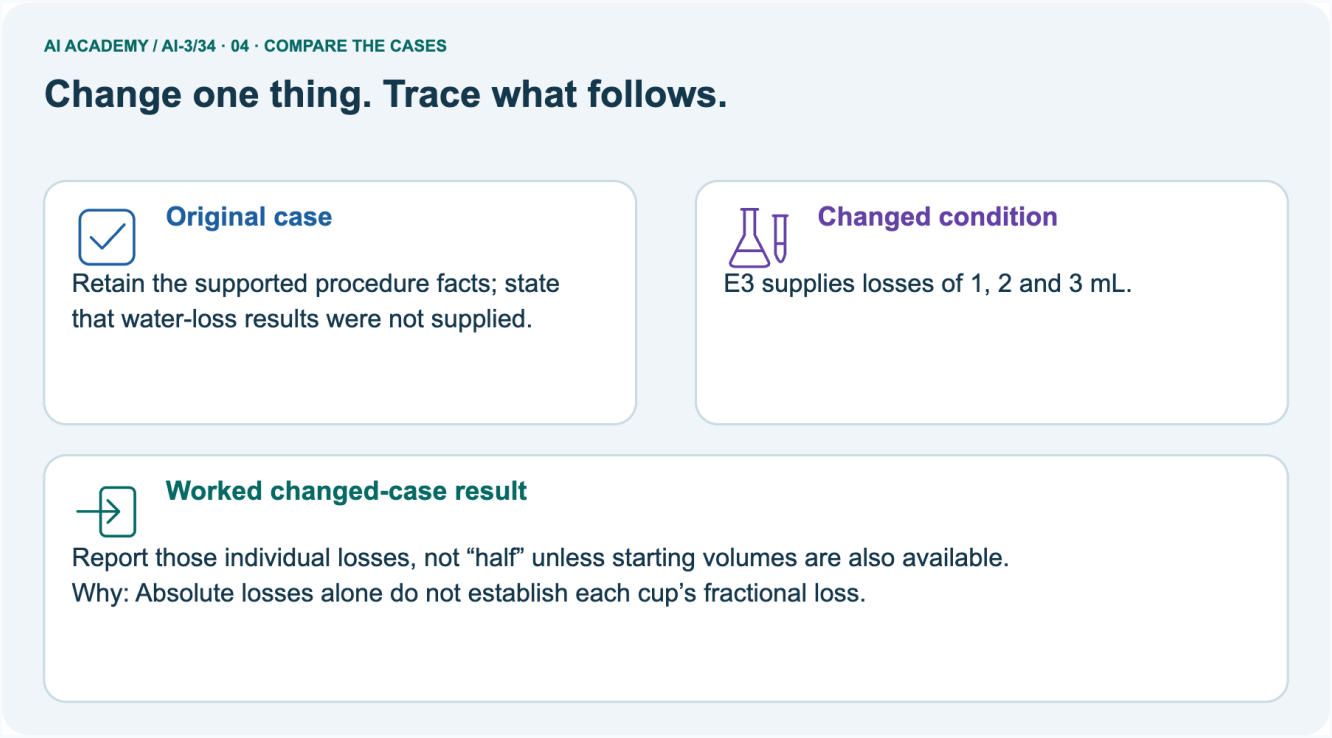
Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



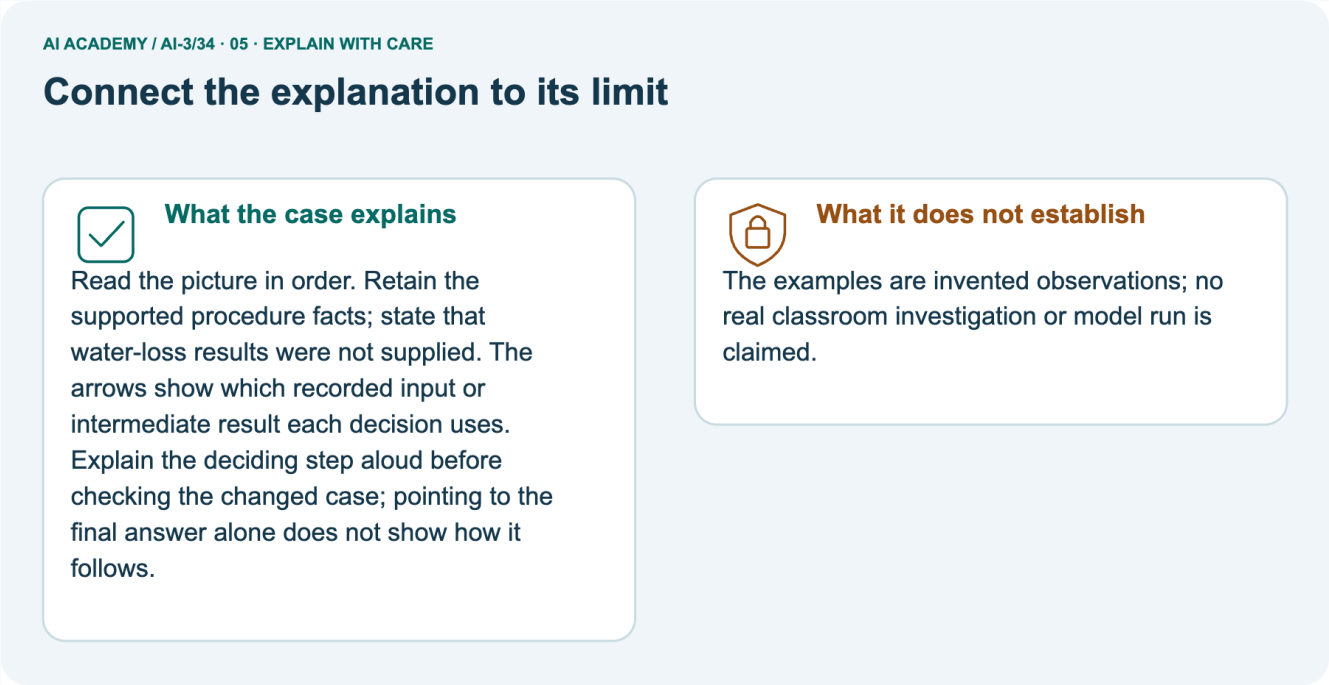
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retain the supported procedure facts; state that water-loss results were not supplied. Changed condition: E3 supplies losses of 1, 2 and 3 mL.

Expected result: Report those individual losses, not “half” unless starting volumes are also available. Reason: Absolute losses alone do not establish each cup’s fractional loss.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

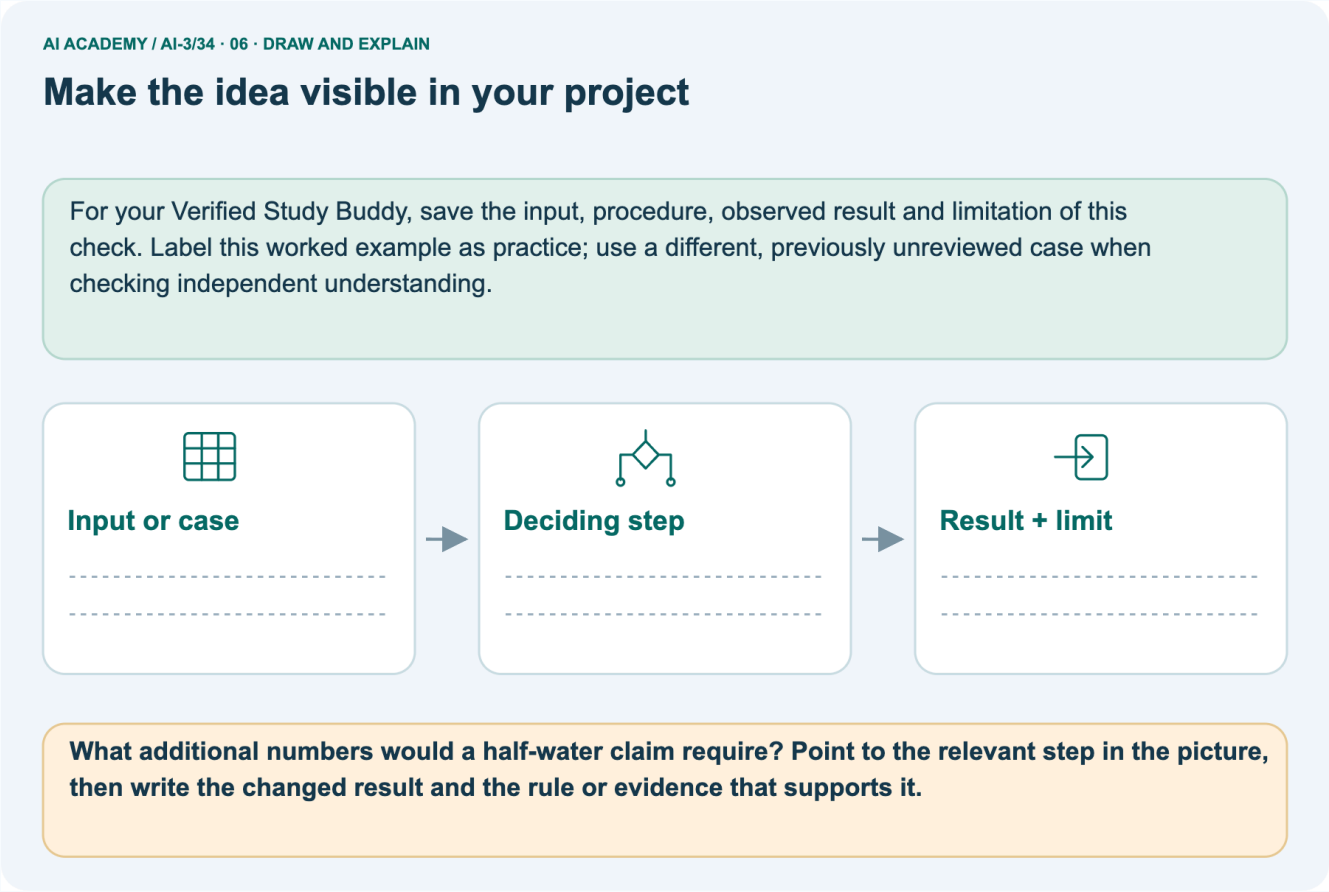
Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The examples are invented **observation**s; no real classroom investigation or model run is claimed.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.